

Introduction

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1 Introduction and Overview of GIS and Software

1.1 Definition of GIS features and functions

A system that creates, manages, stores, visualize and analyzes data and its spatial counterpart is GIS. In another word," A GIS is a computer system capable of capturing, storing, analyzing and displaying geographically referenced information that is data identified according to location. GIS connects data to a map, integrating location data(where things are) with all types of descriptive information(what things are like there). Feature, according to the ESRI GIS dictionary: A representation of a real-world object on a map. The measures and properties of points, lines, and surfaces. In a GIS, geometry is used to represent the spatial component of geographic features. Points define discrete locations of geographic features too small to be depicted as lines or areas, such as well locations, telephone poles, and stream gauges. Functions of GIS: -Data capture: The input of data into a GIS can be achieved through many different methods of gathering. For example aerial photography, scanning, digitizing, GPS or global positioning system is just a few of the ways a GIS user could obtain data. -Data Storage: Some data is stored such as a map in a drawer, while others, such as digital data, can be as a hardcopy, stored on cd or on your hard drive. -Data Manipulation: The digital geographical data can be edited, this allows for many attribute to be added, edited or deleted to the specification of the project. -Query and analysis: GIS was used widely in decision making process for the new commission districts. We use population data to help establish an equal representation of population to area for each district. -Visualization: This represents the ability to display your data, your maps and information.

1.2 Why GIS is important

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1.3 How GIS is applied

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1.4 GIS as an Information System

Geographical Information system plays an important role in development of new technology all over the world. GIS on itself is not a totally new concept but it rather incorporates all well-established methods and techniques of cartography, information technology and programming to provide and display information in a desired manner. From its name, it is evident that it analyzes and displays geographically referenced information. GIS is an information system that is designed to work with data referenced by spatial or geographic coordinates. In other words, a GIS is both a database system with specific capabilities for spatially-referenced data, as well as a set of

operations for working with the data (Star & Estes, 1990). GIS collects data, sorts them, and selects and rebuilds them to find precisely the right piece of information to answer a specific question. The reference to geographic coordinates is an important one, because the coordinates are literally how we are able to link data with the map (Clarke, 1995).

GIS incorporates information from several other techniques such as GNSS and remote sensing. GIS has the capacity to process, manage and manipulate the obtained information, as per the user's command, to obtain output. Development of GIS has led to development on several other sectors like Civil Engineering, Geography, Surveying and so on.

References:

Clarke. (1995). Getting Started With Geographic Information Systems. Prentice Hall.

Star & Estes. (1990). Geographical Information Systems: An Introduction. New Jersey: Prentice Hall.

1.5 GIS and cartography

A map is a graphical representation of an area of land or sea showing physical features, cities, roads, etc. Cartography is the science and art of making and using maps such that the spatial information contained in a map can be communicated effectively. One of the major use of GIS is in the field of cartography.

Before the use of computer based GIS software, cartography was done through scribing which was exhausting and time-consuming. The introduction of GIS has changed the field of cartography. GIS allows digital representations of objects on a map to be connected with a database that describes that object. This ability to directly link data to objects on a map has revolutionized cartography. Geologic maps, because of their complexity, particularly benefit from being produced in a GIS format.

The key trouble among cartography and GIS is that cartography is concerned with illustration whilst GIS is worried with evaluation of spatial relationships. GIS is a product of the improvement of computer-assisted cartography, which generates geo-referenced spatial virtual databases. The databases have become the platform for the introduction of information systems that would be linked, processed, analysed and outcomes displayed within the shape of maps, tables and reports. The structures for those capabilities have become referred to as Geographical Information Systems (GIS). Because of this developmental interconnectivity, Aronoff (1989, p. 103), observes that GIS is often confused with cartographic systems that store maps in automated form. With the help of GIS, anyone can make a digital map from anywhere, this has led to creation of bad maps and in return has diluted the image of cartography.

References:

- <https://geoinfo.nmt.edu/publications/maps/gis/home.html> Silayo, E. H. (2002). Cartography in a GIS Environment. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 34(Part 6), W6.
- Aronoff, S., 1989. Geographic Information Systems: A Management Perspective. WDL Publication, Ottawa, Ontario.
- wiki.gis
- sir's study materials book

1.6 Contributing and allied disciplines

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1.7 GIS data feeds

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1.8 Historical development of GIS.

1.8.1 The birth of GIS

The first example of linking the 'what' with 'where' takes us all the way back to 1854 and a cholera outbreak. At the time, people believed that the disease was being spread through the air. But an enterprising English Doctor, Dr Jon Snow wasn't convinced. So, he decided to map the outbreak locations, the roads and the property boundaries and the water pumps. And, when he did, he made a startling discovery. A pattern emerged. This pattern proved that the disease was not in fact airborne but was being communicated via water and even more specifically by one infected water pump. John Snow's cholera map was a major event connecting the what with the where.

Not only was this the beginning of spatial analysis, it also marked a whole new field of study: Epidemiology, the study of the spread of disease. Snow's work demonstrated that GIS is a problem-solving tool. He put the what on a map to show the where and made a life-saving discovery.

1.8.2 1854 - 1960

During the next hundred years or so, there was limited development in GIS. Mapping was paper based and there was no computer mapping.

By the 1950s, maps were starting to be used in vehicle routing, development planning and locating points of interest.

1.8.3 1960 - 1975

In the period between 1960 to 1975 three major technological advancements in new computer technology led to the birth of modern GIS. Those were: the ability to output map graphics using line printers; advances in data storage and the processing power of mainframe computers. We now had the ability to record coordinates as data inputs and perform calculations on those coordinates.

Roger Tomlinson, widely acclaimed as the 'Father of GIS', during his time with the Canadian Government in the 1960s, was responsible for the creation of the Canadian Geographic Information System (CGIS). CGIS was unique in that it implemented a layering approach to map handling.

The US Census Bureau was also an early adopter of the core principles of GIS. The Bureau began to digitise Census boundaries, roads and urban areas.

The Ordnance Survey GB began to develop their topographic mapping. They used computers to simplify the process of updating future map editions and in 1971, digital mapping was introduced to OS large-scale map

production.

1.8.4 1975 - 1990

This era saw the creation of GIS software. Jack Dangermond, co-founder of Esri Inc had studied environmental science, landscape architecture and urban design. In his own words: "I had some notion of applying computer mapping to my profession". So, in 1967 he headed off to Harvard where he worked in the Laboratory for Computer Graphics. In the mid-1970s, the Laboratory developed the first vector GIS called ODYSSEY GIS. By the late 1970s, the progress in computer memory and improved computer graphic capabilities led to the creation of commercial GIS Software. One of the vendors, was Esri, which is now the largest GIS software company in the world, recognised as a world leading expert in GIS, that has played a key role in the history of GIS.

1.8.5 1990 - 2010

The adoption of GIS into the mainstream took off between 1990-2010. This was facilitated by a number of IT advancements: computers were getting cheaper, faster and more powerful; there were an increasing number of GIS software options and digitised mapping data was more readily available. These advancements, together with the launch of new earth observation satellites and the integration of remote sensing technology with GIS, saw more and more applications being developed. GIS found its way into classrooms, to businesses and to governments across the world.

1.8.6 2010 - 2018

Due to the increased adoption of GIS over the previous twenty years, open source GIS was born. GIS data has become more and more ubiquitous, for instance, Landsat satellite imagery is now accessible to all. GIS is now online, in the cloud, and on your mobile.

1.8.7 Today

Today, Geographic Information Systems (GIS) and spatial analysis are being used by everyone whether we realise or not. Have you noticed that location-based search facility in Google? Do you use an app on your phone to get you from A to B? Do you track your parcel delivery or the taxi you booked? All these examples use spatial information to get you the best answer or the most up to date information.

1.8.8 Other examples include:

- Potential store locations - based on customer locations, customer spending power, route-time analysis, competitor locations.
- Predictive crime analysis - using GIS and spatial analysis resulted in 19 per cent reduction in break ins.
- Real time fleet tracking - reducing delivery costs by 10 per cent.
- Europe's busiest port - 370 vessels per day - helping Dutch infrastructure to rate #1 in the world.
- Air quality - better informed environmental and health policy for Europeans.

Rooted in the science of geography, modern GIS integrates many types of data. It analyses spatial location and organises layers of information into visualisations using maps and 3D scenes. With this unique capability, GIS reveals deeper insights into data, such as patterns, relationships and situations - helping users make smarter decisions.

Hundreds of thousands of organisations in virtually every field are using GIS to make maps that communicate, perform analysis, share information and solve complex problems around the world. This is changing the way the world works.

The Geographic Information System market is predicted to reach US\$9billion by 2024* (*Report by Global Market Insights 2024).

2 GIS and Maps

2.1 Map Projections and Coordinate Systems

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2.2 Maps and their characteristics (selection, abstraction, scale, etc.)

2.2.1 Maps

A map is a symbolic representation of selected characteristics of a place, usually drawn on a flat surface. Maps present information about the world in a simple, visual way. They teach about the world by showing sizes and shapes of countries, locations of features, and distances between places. Maps can show distributions of things over Earth, such as settlement patterns.

2.2.2 Types of Maps

2.2.3 Political Maps

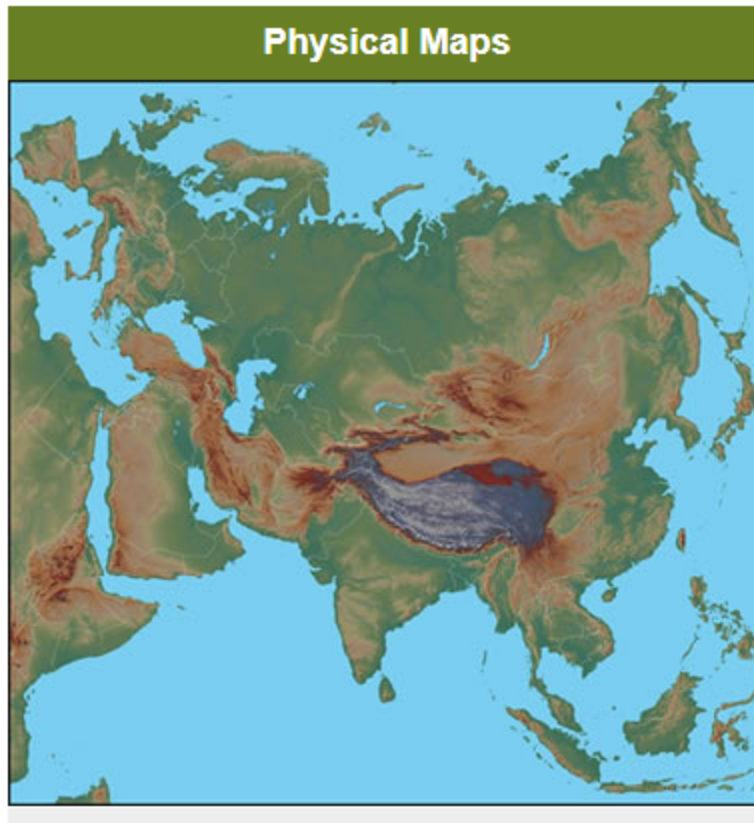
“Political maps” are among the most widely used reference maps. Political maps help people understand the geography of the world. Political maps show how the land is used by people - counties, provinces, countries, town boundaries, etc. They show the geographic boundaries between governmental units such as countries, states, and counties. They show roads, cities and major water features such as oceans, rivers and lakes.



topo_map

2.2.4 Physical maps

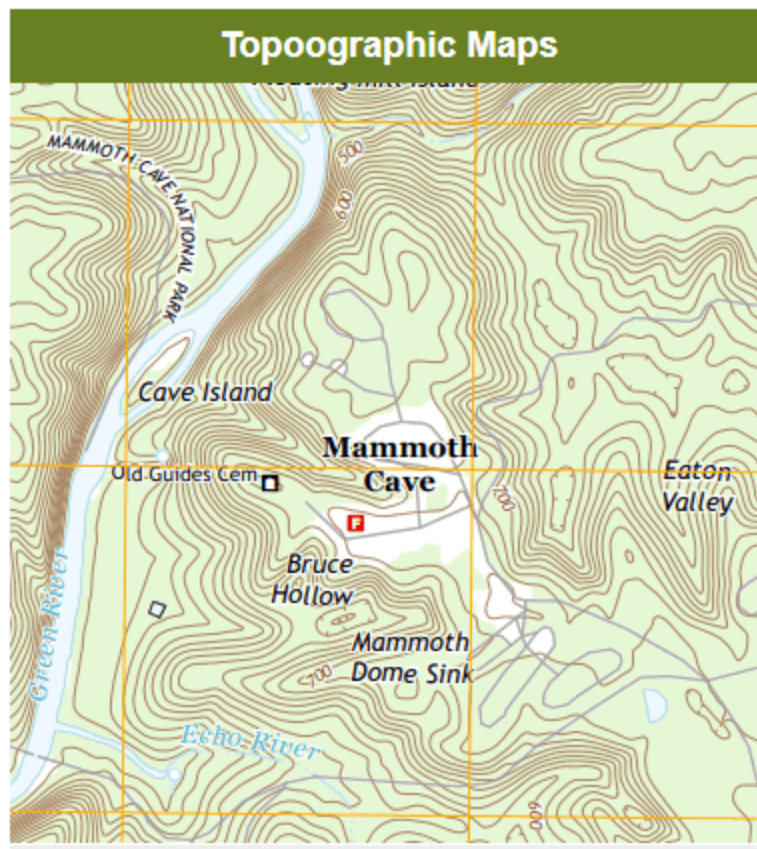
Physical maps are designed to show the natural landscape features of Earth. They are best known for showing topography, either by colors or as shaded relief. Physical maps often have a green to brown to gray color scheme for showing the elevation of the land. Physical maps show the shape of the land - hills, lakes, forests, the coast and so on.



topo_map

2.2.5 Topographic Maps

Topographic maps are reference maps that show the shape of Earth's surface. They usually do this with lines of equal elevation known as "contour lines", but elevation can also be shown using colors (second map), color gradients, shaded relief and a number of other methods. Topographic maps also show other important natural features such as lakes, rivers and streams. Their locations are determined by topography, making them important natural elements of topographic maps.

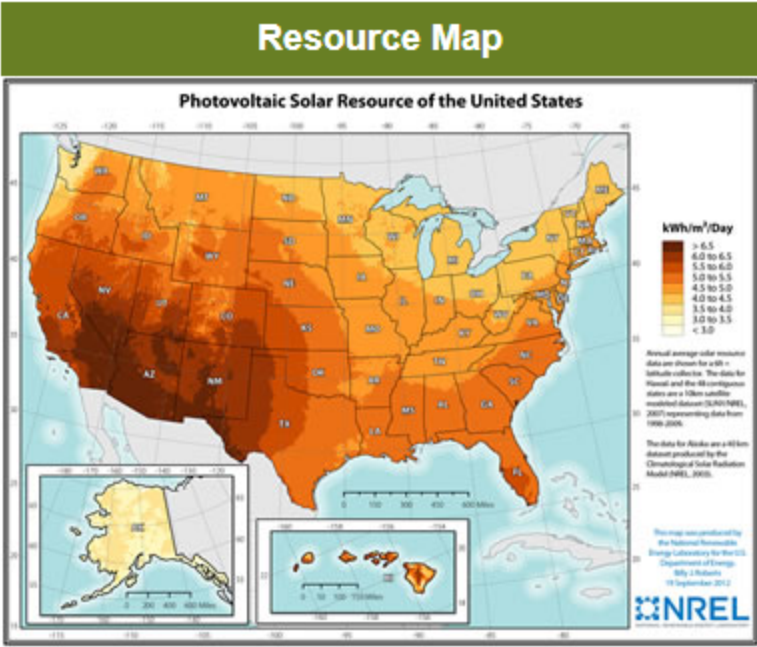


topo_map

fig :-Topographic Map of an area within Mammoth Cave National Park, Kentucky

2.2.6 Resource Maps

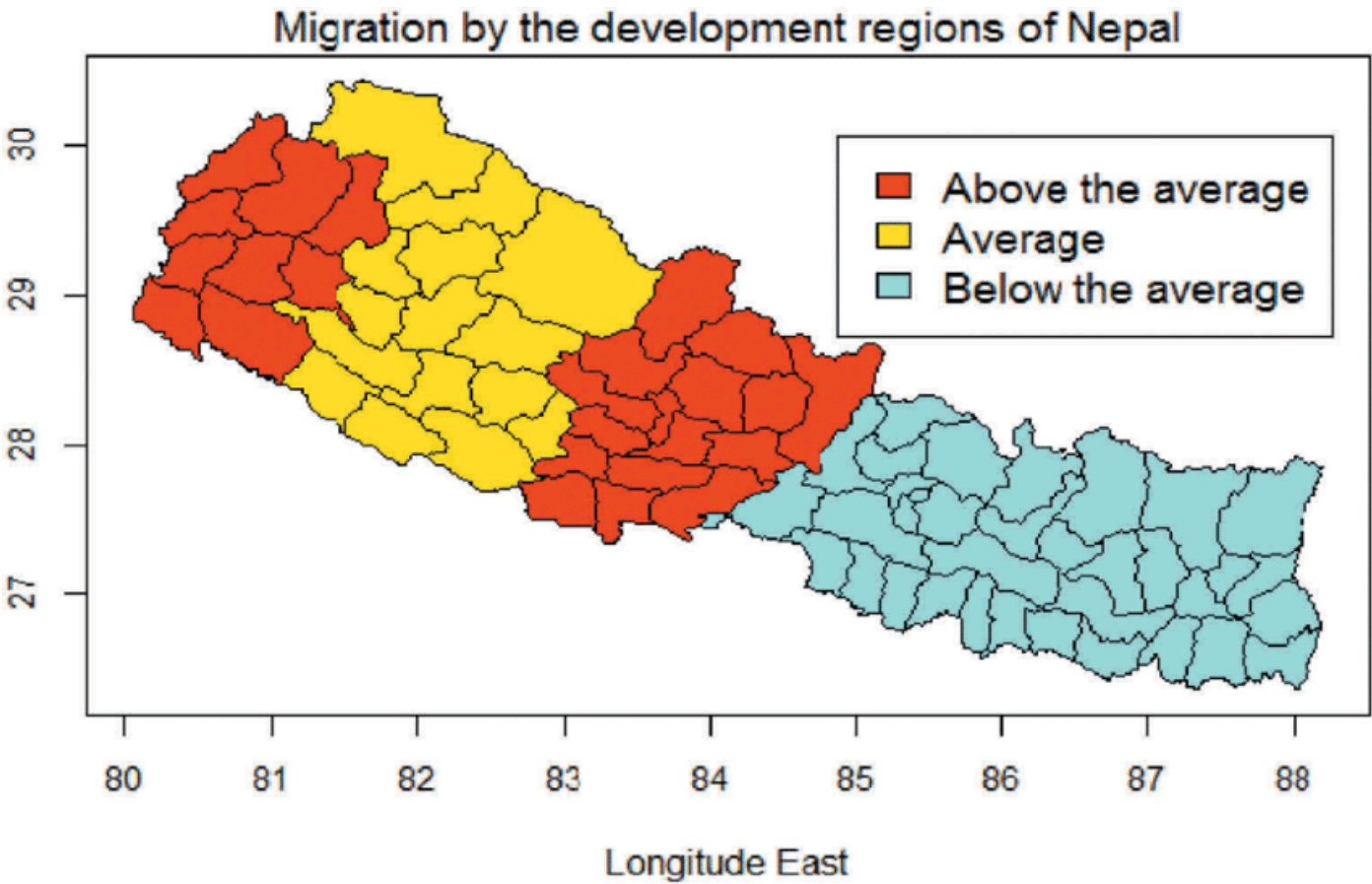
Thematic maps are often made to communicate the geographic distribution of natural resources. These maps might show countries with the highest diamond production or the geographic extent of an oil or gas field. The map shown here illustrates the geographic pattern of solar generating capacity for the United States. Resource maps are important because they help governments understand their natural resource assets and the natural resource assets of their allies and potential enemies. Resource maps help mining companies target their exploration efforts. They are also important for assessing the transportation opportunities and problems associated with the distribution of resources and the location of where they are consumed.



Resource_Maps

2.2.7 Thematic Maps:

Thematic maps display distributions, or patterns, over Earth’s surface. They emphasize one theme, or topic. These themes can include information about people, other organisms, or the land. Examples include crop production, people’s average income, where different languages are spoken, or average annual rainfall.



thematic_Maps

2.2.8 Characteristics of Maps

2.2.9 Selection

The map content represents reduced selections of geographic reality. Maps can depict only those geographic phenomenon which are essential information to readers because it is not possible to display all the geographic phenomena exactly looks like in the Earth's surface. Most maps familiar to us are Orthogonal view of the world. This means that at each point in the map it seems as if one looks straight down to the earth's surface.

2.2.10 Abstraction

Maps are models of reality. They display relations among spatial data, which represent real or abstract geographic phenomena, such as roads or temperatures. Map abstraction refers to the process of explicitly defining and representing real-world features on a map. The three basic geometric forms of geographical features are the point, line, and polygon (or area).

2.2.11 Scale

Map scale is the ratio between the dimensions on the map and on those of reality. All maps are scale models of reality. A map's scale indicates the relationship between the distances on the map and the actual distances on Earth. This relationship can be expressed by a graphic scale, a verbal scale, or a representative fraction. The most common type of graphic scale looks like a ruler. Also called a bar scale, it is simply a horizontal line marked off in miles, kilometers, or some other unit measuring distance. The verbal scale is a sentence that relates distance on the map to distance on Earth. For example, a verbal scale might say, "one centimeter represents one kilometer" or "one inch represents eight miles." The representative fraction does not have specific units. It is shown as a fraction or ratio—for example, 1/1,000,000 or 1:1,000,000. This means that any given unit of measure on the map is equal to one million of that unit on Earth. So, 1 centimeter on the map represents 1,000,000 centimeters on Earth, or 10 kilometers. One inch on the map represents 1,000,000 inches on Earth, or a little less than 16 miles. The size of the area covered helps determine the scale of a map. A map that shows an area in great detail, such as a street map of a neighborhood, is called a large-scale map because objects on the map are relatively large. A map of a larger area, such as a continent or the world, is called a small-scale map because objects on the map are relatively small. Today, maps are often computerized. Many computerized maps allow the viewer to zoom in and out, changing the scale of the map. A person may begin by looking at the map of an entire city that only shows major roads and then zoom in so that every street in a neighborhood is visible.

2.2.12 Symbolization

Maps use symbols that stand for the features of the real world they portray or Cartographers use symbols to represent geographic features. For example, black dots represent cities, circled stars represent capital cities, and different sorts of lines represent boundaries, roads, highways, and rivers. Colors are often used as symbols. Green is often used for forests, tan for deserts, and blue for water. A map usually has a legend, or key, that gives the scale of the map and explains what the various symbols represent. Some maps show relief, or changes in elevation. A common way to show relief is contour lines, also called topographic lines. These are lines that connect points that have equal elevation. If a map shows a large enough area, contour lines form circles. A group of contour line circles inside one another indicates a change in elevation. As elevation increases, these contour line circles indicate a hill. As elevation decreases, contour line circles indicate a depression in the earth, such as a basin.

2.2.13 Accuracy

Accuracy can be a relative term, but it is always measured in relation to something concrete such as a measurement standard. On a map, accuracy usually refers to whether the features represented on the map are represented in the correct location and are represented as the correct type of feature. The accuracy of a map depends on many things. Many accuracy errors are a result of human error. The symbols of the map may also affect the accuracy of the map. Larger symbols may require that they be placed farther apart so that all of them are visible. However, this has the side-effect of moving the locations of the features away from where they really at.

2.2.14 Resolution

Map resolution refers to how accurately the location and shape of map features can be depicted for a given map scale. Scale affects the resolution. As scale decreases, long narrow features such as streams and roads must be represented as lines and smaller area features as points. The minimum sizes and dimensions are sometimes called Minimum Mapping unit.

2.2.15 Map Projections

Through map projections, the spherical surface of the Earth is transformed into a plane. A map projection is a mathematical relationship of geographical coordinates (j, l) and plane coordinates (x, y). Whatever is the mathematical relationship applied to the transformation process, distortions are inevitable. The distortions of the geometric relationships on the sphere (distances, areas, angles, directions), when projected onto the plane, can be controlled by a suitable choice of projection.

2.3 Automated cartography versus GIS

2.3.1 Introduction

(Adopted from https://wiki.gis.com/wiki/index.php/Automated_Cartography.)

Automated map production workflows can be understood by parsing the term into its constituent words. According to Buckley and Watkins: [1]

- *Automation* is the operation or control of equipment, a process, or a system by a machine rather than by hand.
- *Map production* includes the following steps:
 - Deciding the layers to be included
 - Symbolology of each layer
 - Standardization of map elements such as scalebar, title, labeling map compilation,

Automated cartography follows a *workflow*. A “workflow” is a process and/or procedure in which certain tasks are completed. So, “automated map production workflows” are machine-driven processes that result in the completion of tasks that relate to the compilation, construction, or output of a map product.’

2.3.2 Other consideration for automated map production

- Spatial database management (standardization of spatial database)

- Production of multiple maps (e.g., Map atlas) with same design/ theme (symbolology, labeling, selection of colors, placing of map elements and views) but different extent
- Production of maps with same extent but different themes (Geographical map, geological map, environmental map etc)

2.4 Map projections

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2.5 Coordinate systems

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2.6 Precision and error.

2.6.1 Accuracy

Accuracy in GIS is the degree to which information on a map matches real-world values. Therefore, when we refer to accuracy, we are talking about the quality of data and about the number of errors contained in a certain dataset. Highly accurate data can be very difficult and costly to produce and compile. ### Precision Precision refers to how exactly the description of data is. Precise data may be inaccurate because it may be exactly described but inaccurately gathered. (Maybe the surveyor made a mistake, or the data was recorded wrongly into the database). Precise data may be inaccurate because it may be exactly described but inaccurately gathered. (Maybe the surveyor made a mistake, or the data was recorded wrongly into the database).

2.6.2 Error

It is the measure of the estimated difference between the observed or calculated value of a quantity and its true value. The error involves assessing both the imprecision of data and its inaccuracies.

Types of error: - Positional inaccuracy & imprecision: The accuracy and precision of a map is the function of the scale at which a map was created. This means when we see a point on a map, we have its probable location within a certain area.

- Attribute inaccuracy & imprecision: The non-spatial data linked to location may also be inaccurate or imprecise. Inaccuracies may result from mistakes of many sorts. Non-spatial data can also vary greatly in precision. Precise attribute information describes phenomena in great detail. It may be like:
 - i. Name of the places not spelled correctly
 - ii. Numbers do not represent the actual (e.g., population rounded to 1000)
 - iii. A precise description of a person living at a particular address might include gender, age, income, occupation, level of education, and many other characteristics. An imprecise description might include just income or just gender.

- Temporal inaccuracy & imprecision: The age of data may be another obvious source of error. When data sources are too old, some, or a big part, of the information base may have changed. GIS users should always be mindful when using old data and the lack of currency to that data before using it for contemporary analysis.
- Logical inaccuracy & imprecision: Information stored in a database can be employed illogically during the digitizing process. For example,

Political boundary polygons cannot cross

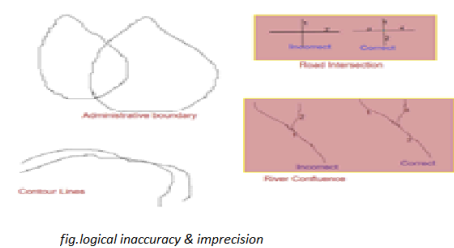
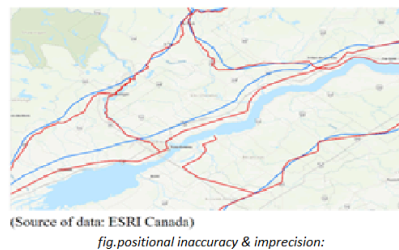
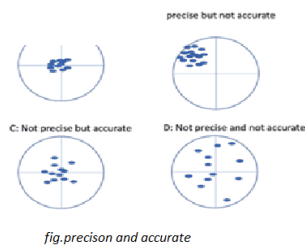
Contour lines cannot meet

Roads always connect at nodes

Rivers connect at the confluence

Some of the logical consistencies are user-defined

- Sources of error may be incompleteness in recorded data, instrumental error, available fund and expertise of the user.
- Logical inaccuracy & imprecision



This is an example image

3 Spatial Data Models

3.1 Concept of data model

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3.2 Raster data model

Raster data models incorporate the use of a grid-cell data structure where the geographic area is divided into cells identified by row and column. This data structure is commonly called raster. While the term raster implies a regularly spaced grid other tessellated data structures do exist in grid based GIS systems. Several tessellated data structures such as hexagonal, triangular and rectangular tessellation exist, however squares are the most commonly used in GIS's. In this model, the data structure involves a division of spatial data into regularly spaced cells. Each cell is of the same shape and size (Buckley, 1997: 28).

The raster cell size is an important factor. Smaller cells improve data quality because they can provide more detail. As cell size increases, data definition decreases or blurs. Cell size in a raster file is referred to as resolution. Spatial resolution of a raster is defined as the size of the smallest recording unit or the smallest size of geographic area represented by a cell.

Conceptually, raster models are the simplest of available spatial data models. We can create a raster of elevation values by encoding each cell with a value that represents the elevation which best represents the elevation in that cell area. When we are finished, every cell will have a value. This will be a numeric value of elevation. We could also give values for other features such as roads, streams, etc., but on separate layers.

A raster is a set of regularly spaced (and contiguous) cells with associated (field) values. The associated values represent cell values, not point values. This means that the value for a cell is assumed to be valid for all locations within the cell (de By and Huisman, 2009: 86).

The size of cells in a tessellated data structure is selected on the basis of the data accuracy and the resolution needed by the user. There is no explicit coding of geographic coordinates required since that is implicit in the layout of the cells. A raster data structure is in fact a matrix where any coordinate can be quickly calculated if the origin point is known, and the size of the grid cells is known. Since grid-cells can be handled as two-dimensional arrays in computer encoding many analytical operations are easy to program. This makes tessellated data structures a popular choice for many GIS software. Topology is not a relevant concept with tessellated structures since adjacency and connectivity are implicit in the location of a particular cell in the data matrix (Buckley, 1997: 28). Since geographic data is rarely distinguished by regularly spaced shapes, cells must be classified as to the most common attribute for the cell. The problem of determining the proper resolution for a particular data layer can be a concern. If one selects too coarse a cell size then data may be overly generalized. If one selects too fine a cell size then too many cells may be created resulting in a large data volumes, slower processing times, and a more cumbersome data set. As well, one can imply an accuracy greater than that of the original data capture process and this may result in some erroneous results during analysis (Buckley, 1997: 28).

As well, since most data is captured in a vector format, e.g. digitizing, data must be converted to the raster data structure. This is called vector-raster conversion. Most GIS software allows the user to define the raster grid (cell) size for vector-raster conversion. It is imperative that the original scale, e.g. accuracy, of the data be known prior to conversion. The accuracy of the data, often referred to as the resolution, should determine the cell size of the output raster map during conversion (Buckley, 1997: 28). Most raster based GIS software requires that the raster cell contain only a single discrete value. Accordingly, a data layer, e.g. forest inventory stands, may be broken down into a series of raster maps, each representing an attribute type, e.g. a species map, a height map, a density map, etc. These are often referred to as one attribute maps. This is in contrast to most conventional vector data models that maintain data as multiple attribute maps, e.g. forest inventory polygons linked to a database table containing all attributes as columns. This basic distinction of raster data storage provides the foundation for quantitative analysis techniques. This is often referred to as raster or map algebra. The use of raster data structures allow for sophisticated mathematical modelling processes while vector based systems are often constrained by the capabilities and language of a relational DBMS (Buckley, 1997: 28).

This difference is the major distinguishing factor between vector and raster based GIS software. It is also important to understand that the selection of a particular 3-3 data structure can provide advantages during the analysis stage. For example, the vector data model does not handle continuous data, e.g. elevation, very well while the raster data model is more ideally suited for this type of analysis. Accordingly, the raster structure does not handle

linear data analysis, e.g. shortest path, very well while vector systems do. It is important for the user to understand that there are certain advantages and disadvantages to each data model.

3.3 Compression

Although the raster data model has many uses in GIS, one of the main operational problems associated with it is the sheer amount of raw data that must be stored. Approximate files sizes are 1.1 MB for a 30 m DEM, 9.9 MB for a 10 m DEM and so on. The memory requirement becomes even higher for high-resolution raster data. Data compression refers to the reduction of data volume, a topic particularly important for data delivery and web mapping. A variety of compression techniques are available for raster data compression. Compression techniques can be lossless or lossy. A lossless compression method allows the original data to be precisely reconstructed. Run-length encoding is an example of lossless compression. A lossy compression method cannot be reconstructed fully to the original form but can achieve high compression ratios. Image degradation through lossy compression can affect GIS related tasks such as extracting ground control points from aerial photographs or satellite images for the purpose of geo-referencing. To improve storage efficiency many types of raster compression technique have been developed such as : run-length encoding, chain encoding, block encoding, wavelet compression , quad-trees encoding. ##### Run-length Encoding (RLE) Run-length encoding is perhaps the simplest compression method and is very widely used. It involves encoding adjacent row cells that have the same value, with a pair of values indicating the number of cells with the same value, and the actual value. Run-length encoding (RLE) allows the cells in each mapping unit to be stored per row for each class (or object) in terms of a beginning cell and end cell, and an attribute (Burrough and McDonnell, 2014). Data type is usually integer. Consider the raster shown in Figure 3-1, it has 8 columns and 8 rows, hence 64 cells. Shaded region has to be stored in RLE by storing the beginning cell number, and ending cell number per row from left to right. ##### Figure:

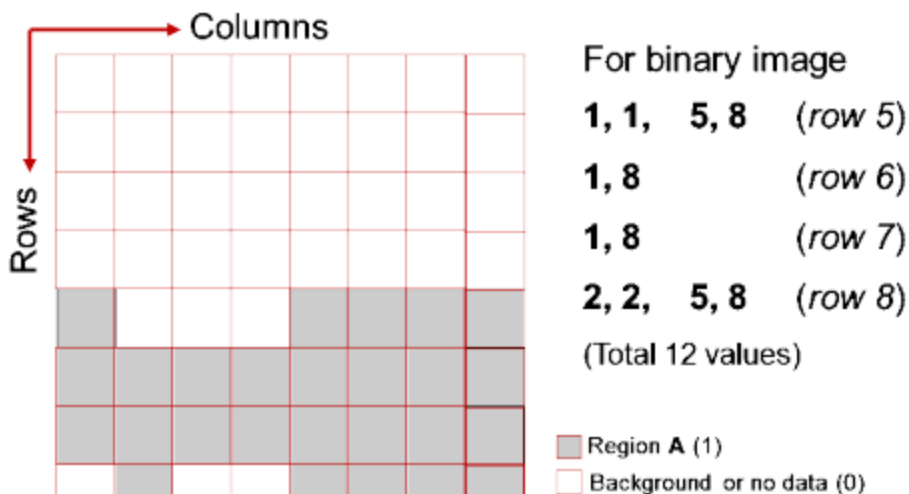


Figure 3-1: RLE of Binary Data (Bhatta, 2014)

In this example, the 26 cells of the shaded region have been completely coded by 12 values. To explain, if we consider the first line, it says that the shaded region starts at 1st cell of row 5 and ends at 1st cell, then it starts again at 5th cell and ends at 8th cell. This example is useful for bitonal (also called binary) raster. For the raster,

having multiple values (or classed) require a different compaction technique to store the data. In this case data can be encoded as a pair of numbers, first the run length and then the cell value as shown in the Figure 3-2.

3.3.0.1 Figure:

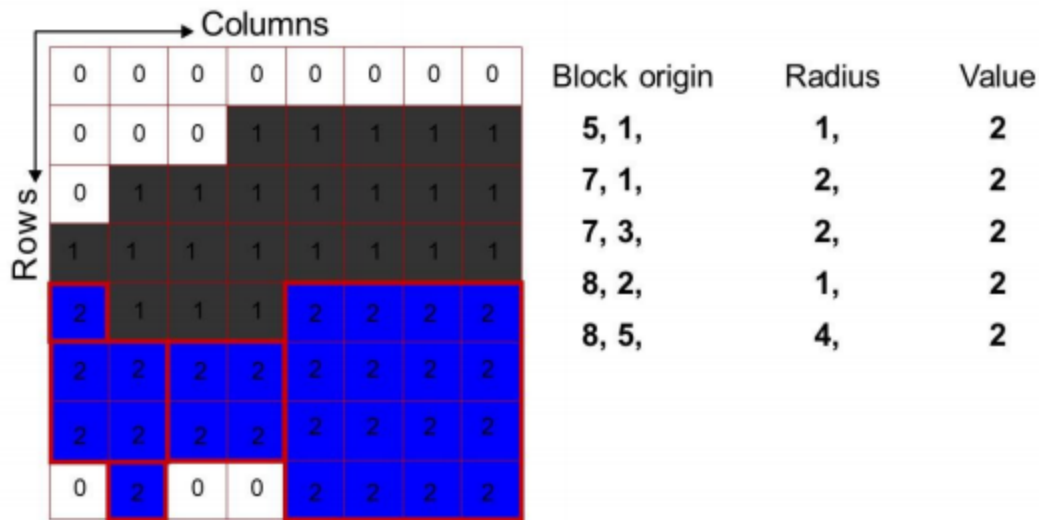


Figure 3-2: RLE of Multi-class Raster (Bhatta, 2014)

This is an example image

3.3.0.2 Chain Encoding

Chain codes represent the raster boundary of a region by giving a starting point and the cardinal direction (east, north, west and south) to follow as we progress around the boundary. Consider the shaded region of Figure 3-1 can be encoded starting from row = 5 and column = 1 in clockwise direction (considering east = 0, north = 1, west = 2, and south = 3): ##### Table

Origin	Value	Sequence of unit
5,1	2	2 ¹ , 5 ⁰ , 2 ¹ , 4 ⁰ , 4 ³ , 4 ² , 2 ¹ , 4 ² , 2 ³ , 2 ¹ , 2 ² , 3 ¹

where the

directions are given by superscript numbers. To describe this structure, from origin (5,1) value 2 can be found in an area of 2 cells in the direction of south, 5 cells in direction of east, 2 cells in the direction of north, and so on. Similarly, other objects (or boundaries) in the raster can also be encoded. Chain codes can be stored using integer data types and therefore provide a very compact way of storing a region representation; they allow certain operations such as estimation of areas and perimeters, or detection of sharp turns, etc. They are also useful for automated raster to vector conversion. Overlay operations such as union and intersection are difficult to perform with chain codes. Another limitation is the redundancy introduced because common boundaries between adjacent regions must be stored twice.

3.3.0.3 Block Encoding

Block encoding is a 2D version of run-length encoding in which areas of common cell values are represented with a single value. An array is defined as a series of square blocks of the largest size possible. Recursively, the array is divided using blocks of smaller and smaller size. It is sometimes described as a quadtree data structure.

3.3.0.4 Wavelet compression

Wavelet compression techniques invoke principles similar to those discussed in the treatment of fractals. They remove information by recursively examining patterns in datasets at different scales, always trying to reproduce a faithful representation of the original. A useful by-product of this for geographic applications is that wavelet-compressed raster layers can be quickly viewed at different scales with appropriate amounts of detail. MrSID (Multiresolution Seamless Image Database) from LizardTech is an example of a wavelet compression technique that is widely used in geographic applications, especially for compressing aerial photographs. Similar wavelet compression algorithms are available from other public and private sources and have been incorporated into the JPEG 2000 standard which is increasingly being used for image compression. Run-length and block encoding both result in lossless compression of raster layers, that is, a layer can be compressed and decompressed without degradation of information. In contrast, the MrSID wavelet compression technique is lossy since information is irrevocably discarded during compression. Although MrSID compression results in very high compression ratios, because information is lost its use is limited to applications that do not need to use the raw digital numbers for processing or analysis. It is not appropriate for compressing DEMs for example, but many organizations use it to compress scanned maps and aerial photographs when access to the original data is not necessary.

3.3.0.5 Quad-tree

The quad-tree method divides a geographic area into square cells using the principles of recursive subdivision of non-homogeneous square array of cells into four equal sized quadrants. The quartering is continued to a suitable level until a square is found to be homogeneous. The underlying structure is a levelled tree where all non-leaf nodes have exactly four descendants. Figure 3-4 shows the successive division of a region into quadrant blocks. This block structure may be described by a tree of degree 4, known as quad-tree. The entire array $2n \times 2n$ cells starts from the root node of the tree, and the height of the tree is at most n levels. Each node has four children, respectively NW, NE, SW, and SE quadrants. Leaf nodes correspond to those quadrants for which no further subdivision is necessary. This structure is efficient for relatively homogeneous areas. Quad-trees are 'variable resolution' arrays in which detail is represented only when available without requiring excessive storage of parts where detail is lacking. Quad-tree representation allows a region to be split up into parts, or to contain holes, without difficulty. The quad-tree has the added advantage of variable resolution. The ease of subsequent processing varies with the data structure used. It is important to realize that quad-tree is more concerned with the overall data model, rather than just saving. ##### Figure:

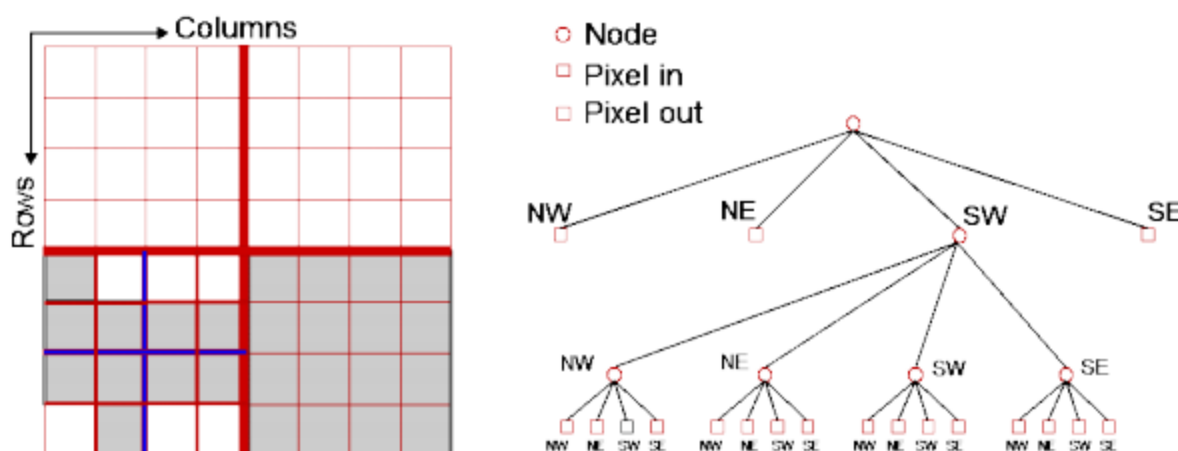


Figure 3-4: Quad-tree Encoding and Quad-tree Hierarchy (Bhatta, 2014)

3.4 Indexing and hierarchical data structures

Indexing: Indexing is a way to optimize the performance of a database by minimizing the number of disk accesses required when a query is processed. It is a data structure technique which is used to quickly locate and access the data in a database. Indexes can be created using some database column. The first column of the database is the search key that contains a copy of the primary key or candidate key of the table.

Search key	Data Reference
------------	----------------

Fig: Structure of Index

The values of the primary key are stored in sorted order so that the corresponding data can be accessed easily. The second column of the database is the data reference. It contains a set of pointers holding the address of the disk block where the value of the particular key can be found.

[This is a link] (<https://www.javatpoint.com/indexing-in-dbms#:~:text=Indexing%20is%20used%20to%20optimize,in%20a%20database%20table%20quickly.>)



This is an example image

A hierarchical database model is a data model in which the data are organized into a tree-like structure. The data are stored as records that are connected to one another through links. A record is a collection of fields, with each field containing only one value. The type of a record defines which fields the record contains.

3.5 Vector data model

Replace this content

3.6 Topology

Replace this content

3.7 TIN data model.

Replace this content

4 Data Sources

4.1 Data Input and Data Quality

4.1.1 Sources of data

4.1.1.1 Spatial Data

- Existing Analog maps
- Aerial photographs
- Satellite image
- Ground survey by GPS ##### Non spatial Data
- Reports
- Tabular information
- Scanned iamges which donnot have geographical reference
- Population census rept
- Health survey
- Agriculture survey reports

4.1.2 Data Quality

4.1.2.1 Data Quality and Components

Quality can simply be defined as the fitness for use for a specific data set. Data that is appropriate for use with one application may not be fit for use with another. It is fully dependant on the scale, accuracy, and extent of the data set, as well as the quality of other data sets to be used. Data quality standards have been developed at both national and international levels in support of mandates for data acquisition and dissemination. Data quality documentation plays a key role in many standards due to the realisation that an understanding of quality is essential to the effective use of geospatial data .

Data Quality Components in SDTS are:

1. Lineage : Refers to source materials, methods of derivation and transformation applied to a database. Includes temporal information (data that the information refers to on the ground).
2. Positional Accuracy : Refers to accuracy of the spatial component . Subdivided into horizontal and vertical accuracy elements. Assessment methods are based on comparison to source, comparison to a standard of

higher accuracy, deductive estimates of internal evidence.

3. Attribute Accuracy : Refers to the accuracy of the thematic component . Specific tests vary as a function of measurement scale. Assessment methods are based on deductive estimates, sampling or map overlay.
4. Logical Consistency : Refers to the fidelity of the relationships encoded in the database. Includes tests of valid values for attributes, and identification of topological inconsistencies based on graphical or specific topological tests.
5. Completeness : Refers to the relationship between database objects and the abstract universe of all such objects. Includes selection criteria, definitions and other mapping rules used to create the database.

4.2 Major data feeds to GIS and their characteristics

Replace this content

4.3 Maps, GPS, images, databases

4.3.1 Maps

- Maps are the representation of part or whole of the Earth's surface onto a plane surface, may it be a digital or a physical surface, with a reduced scale.
- They show all the natural and man-made features existing on the surface with the use of conventional signs and symbols providing the readers with adequate information they desire.
- Essential components of maps are - Title, Legends, Scale, Latitude and Longitude, Direction, Signs and Symbols, Feature texts, etc.
- They are projected on the plane using standard methods of projections
- To create a map, a reference ellipsoidal surface representing the surface of the Earth i.e., a Geoid/datum with Parallels and Meridians is required. Then preferred method of projection is used to project the surface points onto the plane and hence a map can be created.
- Maps are a fundamental part of GIS software. Moreover, they are basic inputs to the GIS software, that enables the user to operate various geoprocessing tools and perform vector and spatial analysis at the region of their interest.
- Various map projections are available in the GIS software and can be used according to one's convenience and requirement.

Types of Maps

1. Topographic maps: These are the general types of maps that depict a section of the Earth's surface and physical features on it – visible landscapes and terrain.
2. Thematic maps: There are used for a specific purpose than topographic maps that depict information that may or may not be directly observable such as population density, forest types, land use, or amount of rainfall. Some examples of thematic maps are category maps, heat maps, cluster maps, etc.

Category map

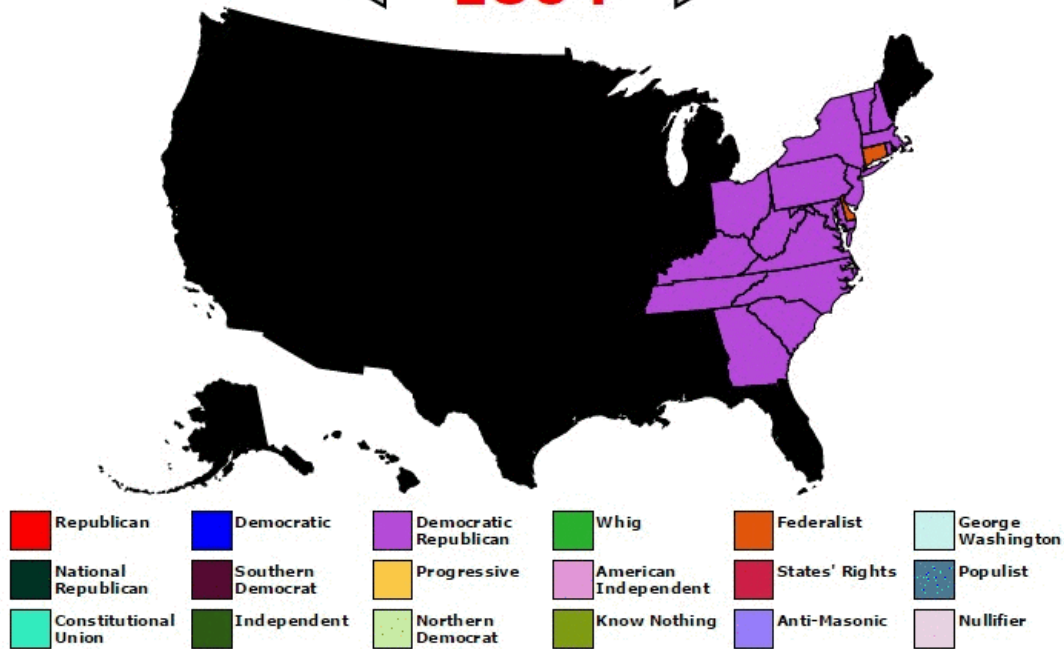
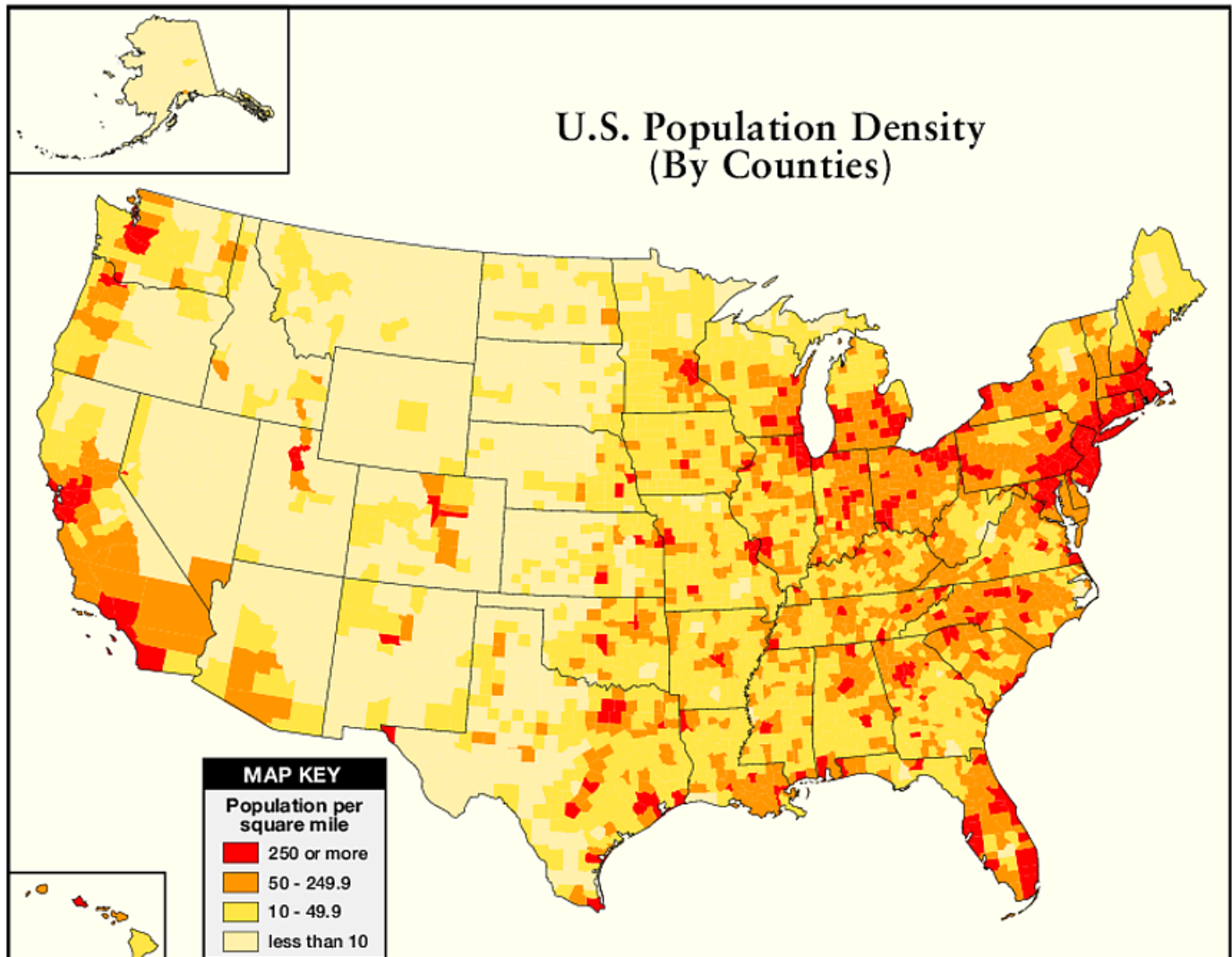
Presidential Election Results by Party◀ **1804** ▶

Image courtesy of

Metrocosm

Heat map



Reference: [This is a link](#)

4.4 Commercial data

Government institutions, consultant agencies and surveying departments collect geographic data for development activities. These organizations sell their data. For instance, Department of Hydrology and Meteorology under Ministry of Energy, Water Resources and Irrigation sells data collected daily at hydrometeorological stations throughout the country.

There are private companies like Maxar Technologies, SkyWatch, OneAtlas, etc. collecting high resolution satellite imageries from their private satellites. The companies sell these imageries to organizations demanding their data.

4.5 Locating and evaluating data

Replace this content

4.6 Data formats

4.6.1 Introduction

GIS is tool for capturing, storing, querying, analyzing and displaying geospatial data. GIS can use different formats of data for different purposes. Data format is a way that geospatial data is stored. And stored data can be manipulated according to our interest. Commonly used data format in GIS are vector data, raster data, CSV(table), CAD etc. ##### 1. Vector Data Format #####1.1 Coverage Data Format Coverage data format. ESRI used coverage data formats, which is built in topology. It uses split system to store geometry and attributes. #####1.2 Shape file: - Shape file is standard non topological data format used in Esri products. -There is no special relationship initially in this data format. #####1.3 Triangulated irregular network.TIN ##### 2. Raster data format. ##### Examples: - Portable network graphics PNG. - Joint Photographic Experts Group JPEG 2000. - TIFF Tagged image file format. - GDAL Virtual format: Allows virtual data to be derived from other data sets that GDAL library can read. - Geotiff. Geo tagged image file format. - Digital elevation model DEM.

4.7 Data quality

Replace this content

4.8 Metadata.

Replace this content

5 Database Concepts

5.1 Database concepts and components

A database is a collection of related information that permits the entry, storage, input, output, and organization of data. A database is an organized collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS). They support electronic storage and manipulation of data. Databases make data management easy. Database management systems help users share data quickly, effectively, and securely across an organization. By providing quick solutions to database queries, a data management system enables faster access to more accurate data.

5.1.1 Components of database:

1. **Hardware:** Hardware is the computer on which a GIS operates. Today, GIS software runs on a wide range of hardware types, from centralized computer servers to desktop computers used in stand-alone or networked configurations.
2. **Software:** This is the set of programs used to control and manage the overall database. Software is the actual DBMS between the physical database and the users of the system. All the requests from the user for accessing the database are handled by DBMS. It is an important component of the database management system.
3. **Data:** Data, in the context of databases, refers to all the single items that are stored in a database, either individually or as a set. Data in a database is primarily stored in database tables, which are organized into columns that dictate the data types stored therein. Data in DBMSs can be arranged hierarchically or in a navigation format. Data is stored in rows and columns in tables that are used by RDBMS. Any information can be easily extracted by using the column name and the row index. DBMS can only be used by one user. The data storage in DBMS is done in the form of a file.
4. **Procedures:** Following steps are used to access the data according to the need: Define Your Needs Create a Shortlist Evaluate Your Options Implement Your Software Maintain Your Database.
5. **Database access language:** Database Languages are the set of statements, that are used to define and manipulate a database. A Database language has Data Definition Language (DDL), which is used to construct a database & it has Data Manipulation Language (DML), which is used to access a database. SQL(Structured Query Language) language allows to handle the information using tables and shows a language to query these tables and other objects related (views, functions, procedures, etc.).

5.2 Flat files

Replace this content

5.3 Relational database systems

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5.4 Data modeling

Data model can be defined as a model that organizes elements of data and standardizes how they relate to one another and to the properties of real-world entities. Hence, data modeling is the process of creating a data model

for an information system by applying certain techniques. They are fundamentals to any database management system. They define how data is connected to each other and how they are processed and stored in system. The most widely used data model are:

5.4.1 Hierarchical Model:

In a hierarchical file structure (tree-structure) there is more than one type of record held in different files. Pointers allow a 'one-to-many' relationship where each master record can have one or more detailed records associated with it. Access to the data is limited to one type of record — the master record. In order to remember this structure, the master record may be thought of as the 'parent' which can be associated with any number of detailed records or 'children' through internally assigned pointers. (Cho, 1995)

A tree structure is followed. Repeating information are allowed in child data segment. A set of field values are attached with data. These record types are same as that of tables in relational model with individual record equivalent to row. A child segment can only have one parent whereas a parent segment may have numerous children.

5.4.2 Network Model:

This model solves the issue of hierarchical model by allowing 'many to many' relationships in the data. Relationships are made through the common attributes the real-world entities shared. If a change is to occur in real world, the model keeps track and acts accordingly. A set consists of an owner and a member record type. An owner can be a member in another set.

5.4.3 Relational Model:

Most GIS use relational model. In this model data and relations are stored neatly on a table. Such formed table has characteristics like:

- Values are atomic (smallest unit possible)
- Each row is unique
- Column values are of same kind
- Each column has a unique name
- Sequence of rows and columns is insignificant

Certain fields are designated as 'keys', which are later used for indexing to save up time.

5.4.4 Object-Oriented Model:

Object oriented data model is based upon real world situations. These situations are represented as objects, with different attributes. All these objects have multiple relationships between them.

5.4.4.1 Elements of Object-oriented data model:

- Objects: The real-world entities and situations are represented as objects in the Object-oriented database model.
- Attributes and Method: Every object has certain characteristics. These are represented using Attributes. The behavior of the objects is represented using Methods.

- Class: Similar attributes and methods are grouped together using a class. An object can be called as an instance of the class.
- Inheritance: A new class can be derived from the original class. The derived class contains attributes and methods of the original class as well as its own.

Bibliography

Cho, G. (1995). A Self-Teaching Student's Manual for Geographic Information . Canberra: University of Canberra and Committee for the Advancement of University Teaching.

5.5 Views of the database

A database view is a subset of a database that is based on a query that runs on one or more database tables. Database views are saved in the database as named queries and can be used to save complete queries that are frequently used. A view can join information from several tables together, or we can say that Views are useful for hiding unwanted information. Most users interact with the database using the database views. A key to creating a useful database is a well-chosen set of views. Luckily, while views are powerful, they are also easy to create. Create custom views of a database to organize, filter and sort records. Database views allow you to easily reduce the complexity of the end user experience and limit their ability to access data contained in database tables by limiting the data presented to the end user. Essentially, a view uses the results of a database query to dynamically populate the contents of an artificial database table. ### Types of database views: 1. Dynamic views: Dynamic views contain data from one or two tables and automatically include all columns of the specified table or tables. Dynamic views can be updated dynamically when related or extended objects are created or modified. 1. Static views: Static views contain data from several tables and the necessary columns of these tables must be specified in the SELECT and WHERE clauses of the static view. Dynamic views can be updated manually when related objects or extended objects are created or modified. ### Importance of views 1. Focus on the data that interests them and on the tasks for which they are responsible. Data that is not of interest to a user can be left out of the view. 1. Define frequently used joins, projections, and selections as views so that users do not have to specify all the conditions and qualifications each time an operation is performed on that data. 1. Display different data for different users, even when they are using the same data at the same time. This advantage is particularly important when users of many different interests and skill levels share the same database.

5.6 Normalization

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5.7 Databases and GIS.

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6 Vector Analysis

6.1 Data management functions

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6.2 Data analysis functions.

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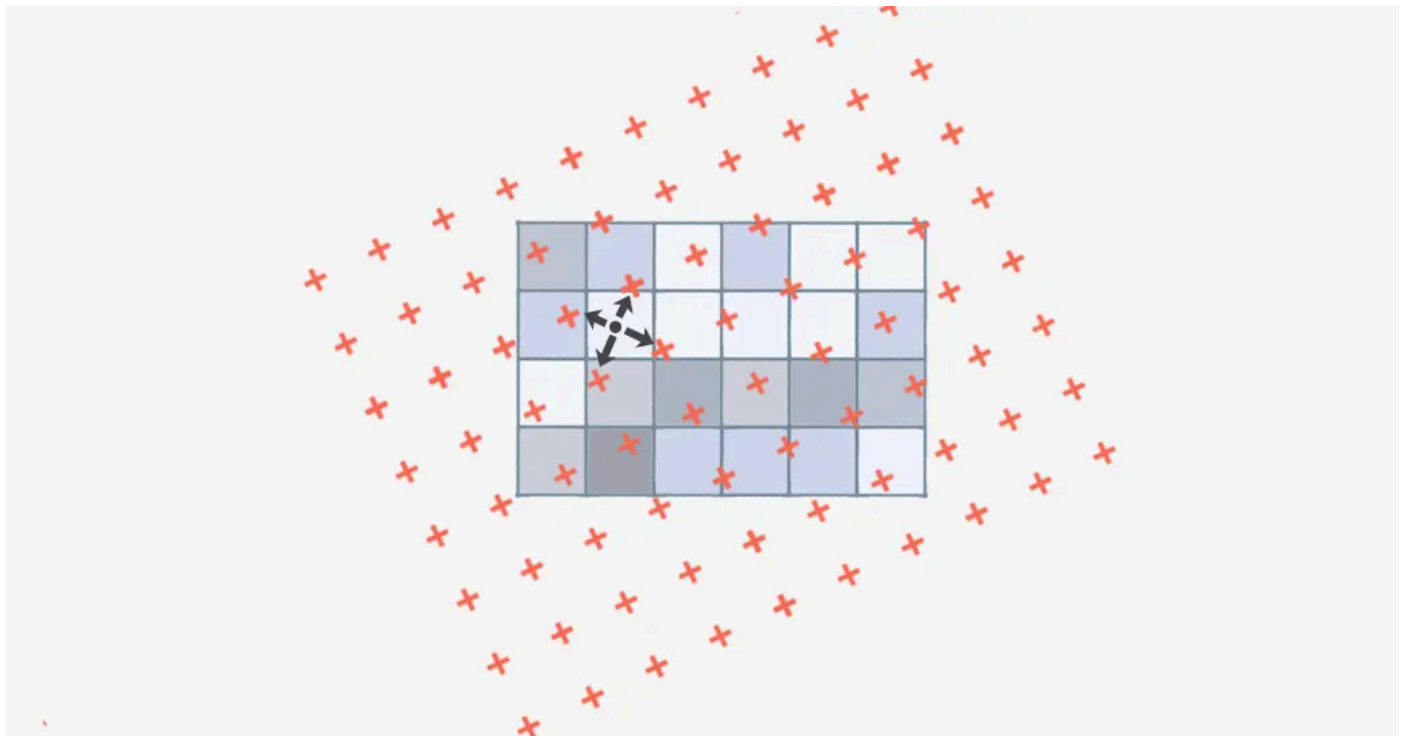
7 Spatial Analysis

7.1 Spatial interpolation methods

7.1.1 Spatial Interpolation

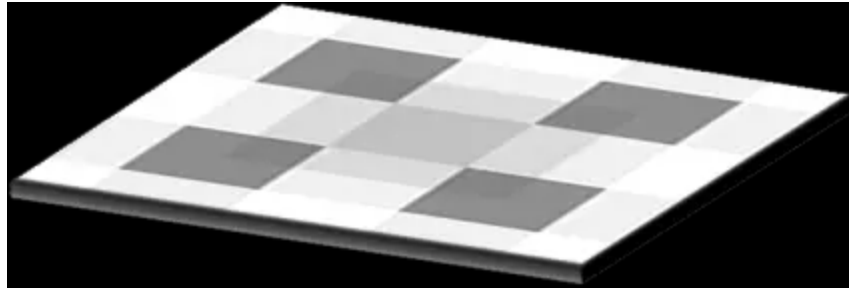
Spatial interpolation is the process of using points with known values to estimate values at other unknown points. In GIS applications, spatial interpolation is typically applied to a raster with estimates made for all cells. Spatial interpolation is therefore a means of creating surface data from sample points. It can be used to predict unknown values for any geographic point data, such as elevation, rainfall, chemical concentrations, noise levels, and so on. For example, to make a precipitation (rainfall) map for your country, you will not find enough evenly spread weather stations to cover the entire region.

7.1.2 Bilinear interpolation



bilinear interpolation

Bilinear interpolation is a technique for calculating values of a grid location based on nearby grid cells. The key difference is that it uses the four closest cell centers to generate an output surface. Using the four nearest neighboring cells, bilinear interpolation assigns the output cell value by taking the weighted average. It applies weights based on the distance of the four nearest cell centers smoothing the output raster grid.

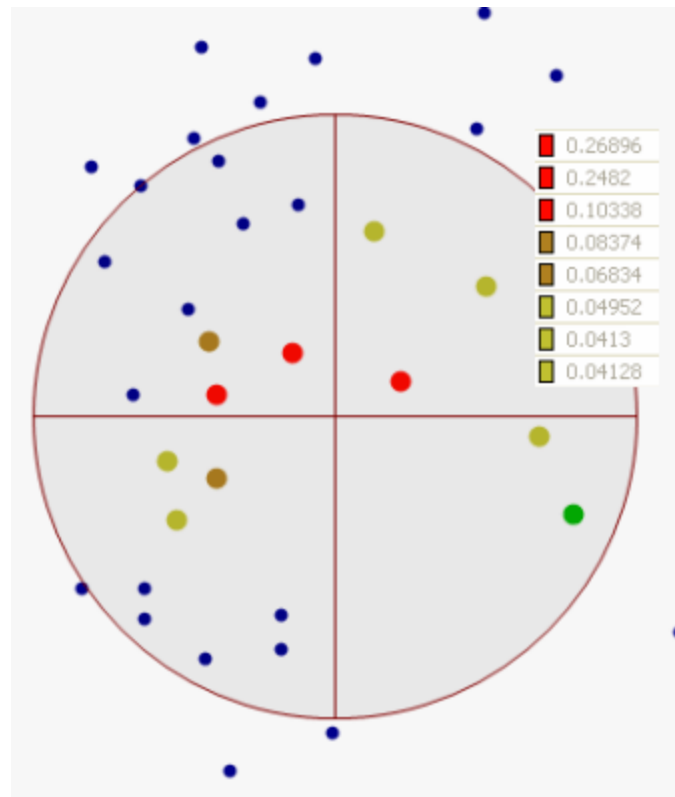


bilinear interpolation

It's recommended to use bilinear interpolation for continuous data sets without distinct boundaries. The surface must be continuous and the closest points must be related. When you run the process, it generates a smoother surface, but not as severe as cubic convolution which uses 16 neighboring cells. The output raster will take only four closest cell centers and apply an average using distance.

7.1.3 Inverse distance weighted (IDW)

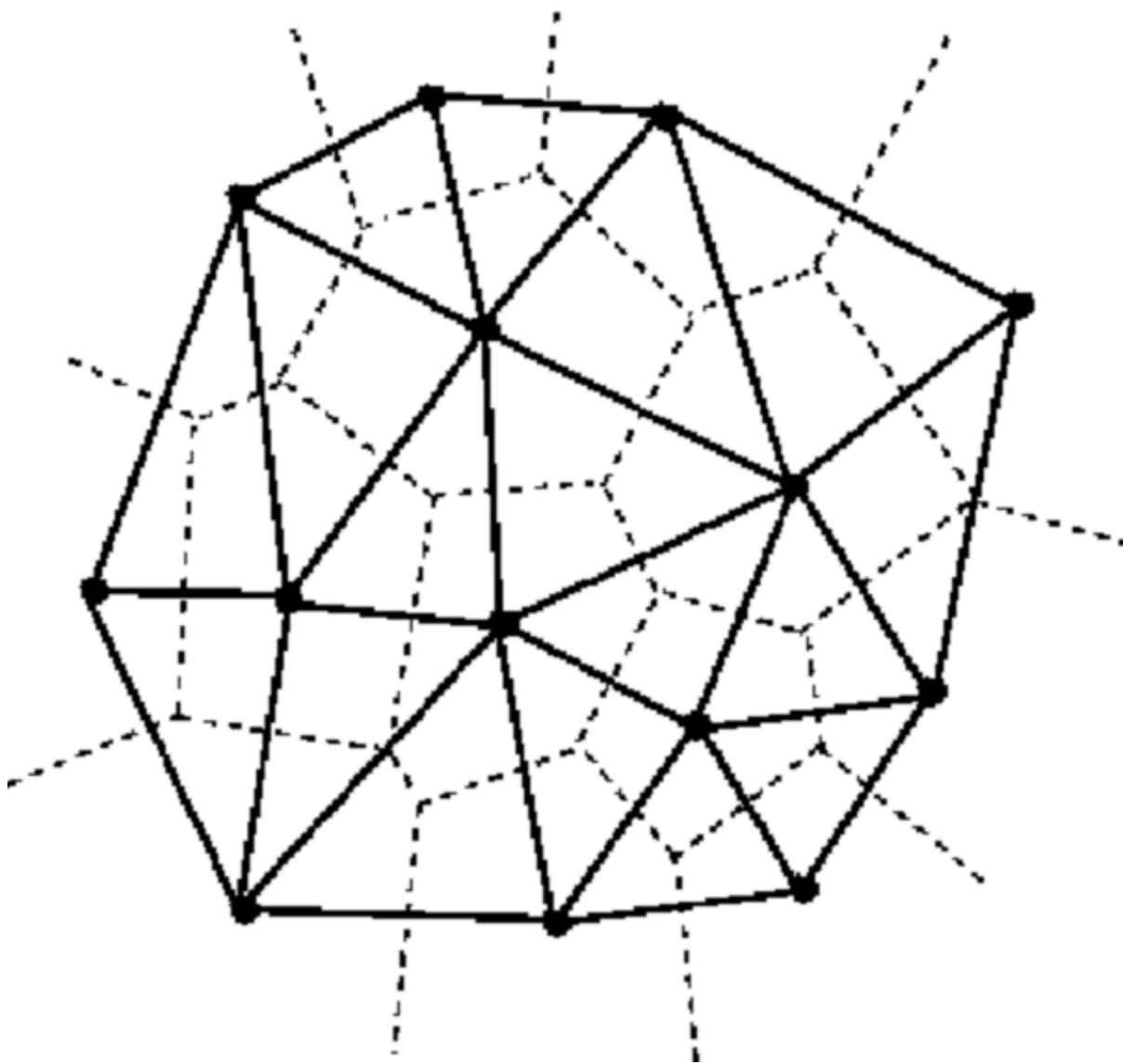
Inverse distance weighted (IDW) interpolation explicitly makes the assumption that things that are close to one another are more alike than those that are farther apart. To predict a value for any unmeasured location, IDW uses the measured values surrounding the prediction location. The measured values closest to the prediction location have more influence on the predicted value than those farther away. IDW assumes that each measured point has a local influence that diminishes with distance. It gives greater weights to points closest to the prediction location, and the weights diminish as a function of distance, hence the name inverse distance weighted. Weights assigned to data points are illustrated in the following example:



IDW

7.1.4 TIN : Triangular Interpolation Network

The TIN technique is maybe one of the most simple spatial interpolation technique. This approach relies on the construction of a triangular network (a specific-case of tessellation) based on the samples spatial location. Multiple triangulation methods might be used to create the network but that of Delaunay is the most commonly reported (fig 2). This method aims at creating non-overlapping triangles (as equilateral as possible) whose circumscribed circle contains only the three points that gave birth to the triangle. Once the network is operational, each new position $z(i)$ where an attribute value needs to be predicted is associated to the triangle T in which it lies. The attribute value at $z(i)$ is calculated as a weighing of the values of the three apexes of the triangle T



TIN

Figure 2. Creation of a triangular network : the Delaunay triangulation

It is clear that the accuracy of the method is strongly dependent on the number of observations within the field, and more especially about the density of observations. When samples are clustered, neighbouring observations will be given a relatively good prediction. On the contrary, for isolated samples, the constructed triangle will be much larger and the prediction will be even more imprecise or flawed. Remind that each predicted value only depends on the three apexes of a triangle. This method might be problematical when small scale variations are present. For instance, the interpolation of elevation values by the TIN technique might generate some discontinuous areas due to abrupt elevation variations (discontinuities, slope) which are not realistic.

Effectively, this interpolation technique does not smooth the data. The TIN technique is available on many softwares and is really efficient from a computation time point of view. It might be applied when lots of observations are available without requiring important memory storage. Note that the network that has been created is also interesting to provide an indication of the influence area of each observation. In that case, other tessellations such as the Thiessen polygons are of particular interest.

7.1.5 Kriging

Kriging is a geostatistical interpolation technique that considers both the distance and the degree of variation between known data points when estimating values in unknown areas. Kriging Interpolation is statistically more sophisticated because it allows identifying distortions in the data, i.e., possible errors on the observations throughout the variance output.

7.2 Raster analysis including topological overlay

7.2.1 Simple mathematical operations



Image

(Figure **a** and **b** shows single input layer and multiple input layer respectively)

7.2.2 Basic operators

- Arithmetic operations (+, -, * etc)
- Relational (>, >=, =)
- Boolean (OR, AND etc)
- Assignment (=)
- Accumulative etc.

7.2.3 Raster analysis -> Map algebra

Algebraic expressions are used to create new raster data from existing ones. This process is also termed as *Map algebra*. There are different types of raster analysis functions as explained below.

7.2.3.1 Local functions

Output raster is created from single or multiple input layers. Corresponding cell from each input layer is used to calculate the value of the output raster for the cell at the same location.

7.2.3.2 Neighborhood (Focal) functions

Value of output raster cell is calculated from the corresponding cell of input raster(s) based on the values of surrounding cells.

Examples:

- Slope: Slope involves the calculation of elevation difference between any given cell with all the surrounding cells and dividing by the distance. Calculation of elevation difference and distance is followed by calculation of slope with respect to each surrounding cell and the selecting the highest value.
- Aspect: Aspect is the direction of the maximum slope which also uses the neighboring cells
- Image filters: e.g., low pass filter, high pass filter etc

7.2.3.3 Zonal functions

- Raster data is divided into zones and the output contains derived value for each zone. Zonal statistics is one of the most used zonal functions used in raster analysis.

7.2.3.4 Global functions

- Global (related to whole the raster data) mean, median mode etc are the examples of global raster functions.

Following image illustrates each of the above functions.  Image of map algebra function types

7.3 Map calculations

Replace this content

7.4 Statistics

Replace this content

7.5 Integrated spatial analysis.

Replace this content

8 Surface Model

8.1 DEM

Replace this content

8.2 Slope

Replace this content

8.3 Aspect

Replace this content

8.4 Other raster functions.

Various raster functions available in GIS are: 1. Slope: A slope is formally described by a plane at a tangent to a point on the surface. The slope has two components:

- i. Gradient: The maximum rate of change of the elevation of the plane

ii. Aspect: The direction of the plane with respect to some arbitrary zero (usually North)

2. Curvature:

There are mainly two curvatures.

i. Profile Curvature: Rate of change of gradient.

a. Convex: Increase in slope gradient

b. Concave: Decrease in gradient in downslope

c. Planar: Constant Gradient

ii. Plan Curvature: Rate of change of gradient.

a. Convex: Aspect lines diverse downslope

b. Concave: Aspect lines converge downslope

c. Planar: Aspect constant

9 River Network Generation

9.1 Flow direction

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9.2 Flow accumulation

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9.3 River network and watershed boundary delineation.

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10 GPS

10.1 Basic concept of GPS

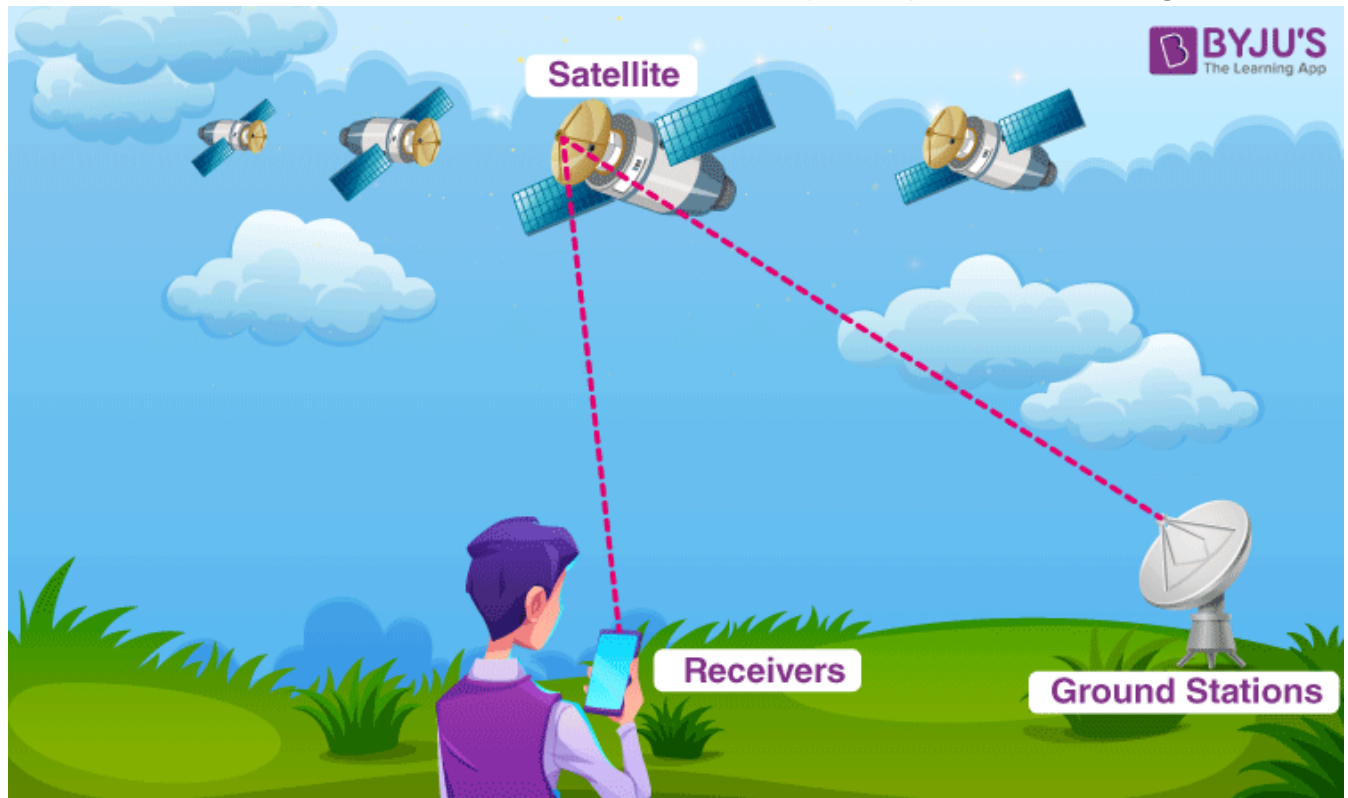
10.1.1 What is GPS?

- GPS stands for Global Positioning System.
- It is a radio navigation system used in land, sea, and air to determine the exact location, time and velocity irrespective of weather conditions. The US military first used it in the year 1960.

10.1.2 Components of a GPS system

GPS is a system and it is made up of three parts: satellites, ground stations, and receivers. Following are the functionalities of each of these parts:

1. Satellites act like the stars in constellations, and we know where they are because they invariably send out signals.
2. The ground stations make use of the radar to make sure the satellites are actually where we think they are.
3. A receiver is a device that you might find in your phone or in your car and it constantly seeks for the signals from the satellites. The receiver figures out how far away they are from some of them. Once the receiver calculates its distance from four or more satellites, it knows exactly where you are. ### The image



10.2 How GPS works

Replace this content

10.3 DGPS

A Differential Global Positioning System (DGPS) is an enhancement to the Global Positioning System (GPS) which provides improved location accuracy and better signal. DGPS uses a fixed, known position to adjust real-time GPS signals to eliminate pseudo-range errors. An important point to note is that DGPS corrections improve the accuracy of position data only and have no effect on results that are based on speed data.

GPS provides an accuracy of about 10 meters, but DGPS can provide accuracy around 1 meter, even beyond that of 10 cm.

10.3.1 Cause of Pseudo-range Delay:

GPS signals coming from satellites down to the ground have to travel through layers of the earth's atmosphere, so they are subjected to delays. This affects the time taken for the signal to travel from any given satellite to a GPS receiver, which introduces slight error into the GPS engine, causing an error in the measured position.

First, the signals have to travel through the **ionosphere**, which is the outer edges of the atmosphere. This layer of the atmosphere is hit by solar radiation, which causes particles to split up and become positively charged. The ionosphere has the largest impact on electromagnetic signals passing through, and hence affects the radio signals coming down from satellites too. A relatively large delay is encountered in this phase. The actual number depends on a couple of factors like receiver location, satellite location, time of day, solar flare activity, etc. Considering these factors, a delay of up to 16 ns is estimated for the signal passing through the layer. This is a rough figure and is changing all the time. This can introduce up to a 5 m error to the captured position.

The second layer the GPS signals travel through is the **troposphere**. This is the 'weather' section of the atmosphere, so includes conditions such as clouds, rain, and lightning. This adds a much smaller delay to the signal of up to 1.5 ns, which can introduce up to a 0.5 m positional error.

These delays are random delays that fluctuate. Each delay is also specific to every individual satellite, as they are positioned in different areas around the world, so their signals will be subjected to different atmospheric conditions.

Reference: [This is a link](#)

10.4 Errors in GPS

GPS deals with lots of errors and their corrections:-

1. Errors due to Atmospheric effects

- Humidity, temperature and pressure variations changes the velocity of light signal in the atomosphere.

10.5 Application.

Replace this content

11 Introduction to Remote Sensing

11.1 Concept of remote sensing

Remote sensing is the art and science of gathering information about material object area without coming into physical contact with them. Information about their location, surface, body can be taken frequently. Information is gathered by using electromagnetic radiation EMR. EMR is form of energy,that transmits and reflect to correlate presence of different body at that location. correlation can be obtained based on variable wavelength of energy. Multi spectral bands can be obtained as we exercised about MODIS data and Sentinel data.

11.2 Electro-magnetic spectrum and windows

Replace this content

11.3 Spectral signature of different land use

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11.4 Introduction to different satellites

Satellite is designed to capture, store, manage, analyze, and visualize all types of geographical data. GIS allows for the integration and collective analysis of geospatial data from multiple sources, including satellite imagery, GPS recordings, and textual attributes associated with a particular space. There are nearly 6,542 satellites orbiting the Earth as of January 1, 2021. Out of which 3,372 satellites are active, and 3,170 satellites are inactive. The main satellites are: Open Data Satellites 1 Landsat. Landsat's incredible long-lived legacy has archived Earth's history for over 40 years 2 Sentinel 3 Terra 4 EnviSAT 5 Corona 6 Earth Observing-1 Mission (EO-1) 7 China-Brazil Earth Resource Satellite (CBERS) 8 Project for On-Board Autonomy (PROBA) Four satellites are needed for GPS because each data from one satellite put you in a sphere around the satellite. By computing the intersections you can narrow the possibilities to a single point. Three satellites intersection places you on two possible points. The last satellite give you the exact location. There are nine different types of satellites i.e. Communications Satellite, Remote Sensing Satellite, Navigation Satellite, LEO, MEO, HEO, GPS, GEOs, Drone Satellite, Ground Satellite, Polar Satellite. Satellite imagery greatly enhances a GIS mapping project and serves as a source of information and data to support analysis and classification for geospatial assessment and modeling. Satellite Images and computers can help geographers improve their knowledge of the world by forming images from what an area looks like from above Earth. They collect information that we cannot see from the planet's surface that helps geographers make accurate maps.

11.5 Resolutions in RS

Replace this content

11.6 Application of remote Sensing.

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12 Making Maps

12.1 Map functions in GIS

This is the final outcome from GIS. This is the graphical representation for readers. It includes:

- Make ready the data and spatial features in appropriate format.
- Select appropriate information to be depicted in the map
- Select map size
- Select the scale
- Symbolization
- Storage
- Update

12.2 Map design

Arthur H. Robinson, an American cartographer influential in thematic cartography, stated that a map not properly designed “will be a cartographic failure.” A map should be visually attractive, while efficiently communicating desired information to an intended audience. A good map should have following characteristics:

12.2.1 Charactersitics of a good map

- Clarity: the map focuses on its purpose with as little distraction as possible.
- Order: the elements of the map are organized to guide the map reader to accomplish the intended purpose.
- Balance: elements are distributed to give the map a feeling of “evenness.”
- Contrast: important elements stand out clearly against less important ones.
- Unity: the various elements come together to make the map appear as a single coherent whole.
- Harmony: the elements all seem to fit together naturally.

12.2.2 Map layout

It is the composition of the main map image with its secondary accessories such that the map is clear, concise and is effective. A poorly composed map is perplexing and should be avoided as it deviates from the true purpose of the map. A good layout of map contains following elements:

12.2.3 Layout

1. Title
2. One or more map images, including inset maps
3. Legend or key
4. Visual or narrative explanation of the map scale
5. Supporting media, such as photographs, diagrams, and text
6. North arrow or other depiction of orientation
7. Metadata, explaining such information as the currency of the information, sources used, projection, copyrights, and authorship

12.2.4 The basics of map design include:

- Symbology : A map conveys its message through the use of different symbols. A map visuals can be categorized by using symbols. Symbols can be used to define various features like for airports, cities etc.
- Color: Colors should be chosen in such a way that it isn't blinding the audience while being distinct enough to clarify the user on the meaning of a map.

- Outlines, borders, and stroke widths These should be chosen in such a way that it fits the context of the map. It should be used to emphasize details.
- Labeling The labels on the map should be clear enough for audience to read the text and have a good feature association such that the reader can easily recognize the feature that the text is referring. The font chosen should look professional, easy to read and lower than the map title's font.

12.3 Map elements

Replace this content

12.4 Choosing a map type

Replace this content

12.5 Exporting map in different format

Replace this content

12.6 Printing a Map

Replace this content

13 Book Editing

13.1 How to edit this book

13.1.1 How to edit this book

- Check which sections you are going to edit at this link: [Link to selected book chapters](#)

13.1.2 Skeleton of the book is available at the following link

[Skeleton of the book](#)

13.1.3 Steps for editing the book

- Click the 'Book Chapter' link on the home page [Book Chapter Selection for you](#)

- Link to the editable sections and image upload will be visible on the above link as shown below

Select following links (Edit Book Chapters)

[Automated cartography versus GIS](#)

[Raster analysis including topological overlay](#)

[How to edit this book](#)

Select following links(Upload Images)

[Automated cartography versus GIS](#)

[Raster analysis including topological overlay](#)

[How to edit this book](#)

- Prepare (upload) the images which you want to put on your book section (**Note: don't use space or '-' or any special characters in the file name except underscore**)
- If you upload an image named 'gis_features.png' for section 10.1 then the file name will be changed to 10.1_gis_features.png.
- Click the link to the sections you want to edit.
- You will see a textarea field which you need to edit.
- After filling the textarea, click submit
- You can edit it multiple times

13.1.4 Adding sub-sections

- Chapter title and section titles are already prepared. You need to start from sub section (3rd level).
- Start Sub-section with three hashes (###) and sub-sub-section with four hashes (####)
- Always mention source when you use one. Link to source can should be like [Link name](Link URL)
- Image can be added by adding a line as ![Image title](/GIS/gisbook/booking/15.1_booksections.png)
- Bullets should start with dash character (-)
- Numbered list should start with '1.'

13.1.5 Numbered list

1. This is line number 1
2. This is line number 2
3. ETC

13.1.6 Testing

- Upload an image for the chapter (for example filename: xxx.png, section: 9.5)

- Copy and paste the following text in the text area and submit

Testing bullet list

- This is an example sentence
- This is bullet list

Testing numbered list

1. This is a numbered list
1. This is another line

The image

![This is an example image](/GIS/gisbook/booking/9.5_xxx.png)

A link

[This is a link](https://mrakhilesh.com)